TASK 2: EXPLORATORY DATA ANALYSIS (EDA)

Course: Data Analysis

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Task: Task 2 – Exploratory Data Analysis

Dataset: Books to Scrape Dataset

Tools and Technologies: Python, Pandas, Matplotlib, Visual Studio Code

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1. Introduction

Exploratory Data Analysis (EDA) is an important step in data analysis that involves examining a dataset to understand its structure, characteristics, patterns, and potential problems before conducting further analysis.

In this task, EDA was performed on the book dataset created during Task 1 using web scraping. The dataset contains 100 book records collected from the first five pages of the Books to Scrape website. It includes book titles, prices, availability status, ratings, and price categories.

Python libraries, particularly Pandas and Matplotlib, were used to inspect the dataset, calculate descriptive statistics, identify missing values and duplicate records, investigate potential outliers, and visualize patterns in the data. Correlation analysis was also performed to explore the relationship between book prices and ratings.

2. Objectives

The main objectives of this task were:

- To ask meaningful questions about the dataset before performing analysis.
- To examine the structure, dimensions, variables, and data types of the dataset.
- To calculate descriptive statistics for numerical variables.
- To identify trends and patterns in book prices, ratings, availability, and price categories.
- To detect missing values, duplicate records, and potential data quality issues.
- To investigate potential outliers in book prices.
- To formulate a hypothesis about the relationship between book prices and ratings.
- To use statistical analysis and visualizations to investigate the hypothesis.
- To summarize the findings and identify issues that may require further investigation.

3 Dataset Description

- The dataset used in this task was generated during Task 1 through web scraping of the Books to Scrape website. The file was loaded into Python using the Pandas library for exploratory analysis.
- The dataset contains 100 records and five original columns:

Variable	Description

Title	Name of the book
Price	Price of the book represented as a numeric value
Availability	Availability status of the book
Rating	Book rating represented by categories from One to Five
Price Category	Derived category based on price: Low, Medium, or High

- For additional analysis, the Rating variable was mapped to a numeric scale from 1 to 5. This numeric representation was used to calculate the correlation between price and rating.

	A	B	C	D	E
1	Title	Price	Availabilit	Rating	Price Cate
2	A Light in the Attic	51.77	In stock	Three	High
3	Tipping the Velvet	53.74	In stock	One	High
4	Soumission	50.1	In stock	One	High
5	Sharp Objects	47.82	In stock	Four	High
6	Sapiens: A Brief History of Humankind	54.23	In stock	Five	High
7	The Requiem Red	22.65	In stock	One	Medium
8	The Dirty Little Secrets of Getting Your Dream Job	33.34	In stock	Four	Medium
9	The Coming Woman: A Novel Based on the Life of the Infamous Fe	17.93	In stock	Three	Low
10	The Boys in the Boat: Nine Americans and Their Epic Quest for Gol	22.6	In stock	Four	Medium
11	The Black Maria	52.15	In stock	One	High
12	Starving Hearts (Triangular Trade Trilogy, #1)	13.99	In stock	Two	Low
13	Shakespeare's Sonnets	20.66	In stock	Four	Medium
14	Set Me Free	17.46	In stock	Five	Low
15	Scott Pilgrim's Precious Little Life (Scott Pilgrim #1)	52.29	In stock	Five	High
16	Rip it Up and Start Again	35.02	In stock	Five	Medium
17	Our Band Could Be Your Life: Scenes from the American Indie Und	57.25	In stock	Three	High
18	Olio	23.88	In stock	One	Medium
19	Mesaerion: The Best Science Fiction Stories 1800-1849	37.59	In stock	One	Medium
20	Libertarianism for Beginners	51.33	In stock	Two	High
21	It's Only the Himalayas	45.17	In stock	Two	High
22	In Her Wake	12.84	In stock	One	Low
23	How Music Works	37.32	In stock	Two	Medium
24	Foolproof Preserving: A Guide to Small Batch Jams, Jellies, Pickles,	30.52	In stock	Three	Medium
25	Chase Me (Paris Nights #2)	25.27	In stock	Five	Medium

4. Questions for Exploratory Data Analysis

Before performing the analysis, the following questions were defined to guide the investigation:

1. How many records and variables are present in the dataset?
2. What are the data types of the available variables?
3. Are there any missing values or duplicate records?

4. What are the average, minimum, maximum, and median book prices?
5. How are book ratings distributed across the dataset?
6. How many books belong to each price category?
7. What does the distribution of book prices reveal about the dataset?
8. Are there any potential outliers in the Price variable?
9. Is there a measurable relationship between book prices and ratings?
10. Are there any data quality issues that should be considered before further analysis?

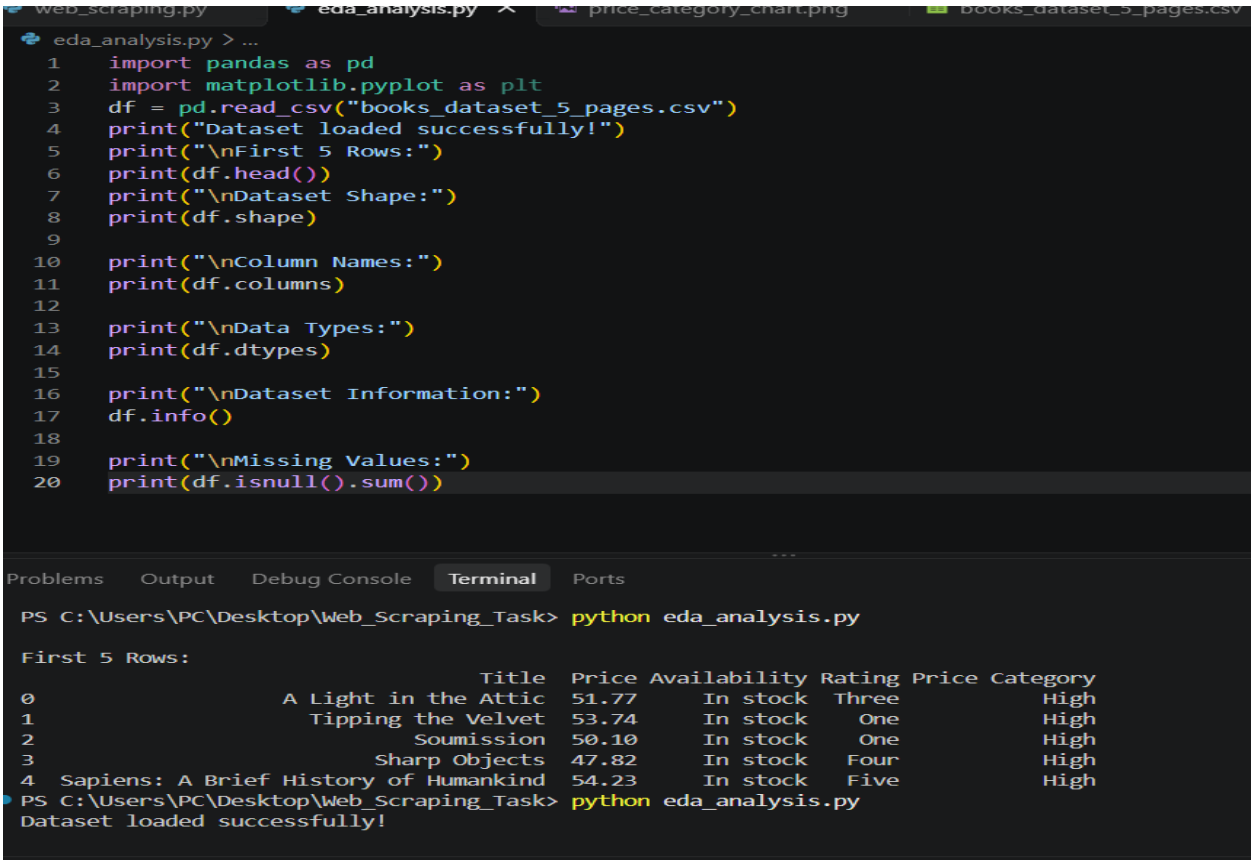
These questions provided a structured approach for exploring the dataset and interpreting its characteristics.

5. Data Structure and Types

The structure of the dataset was examined using Pandas functions such as shape, columns, dtypes, and info().

The dataset contains 100 rows and five original columns. The Price variable is stored as a floating-point numeric value, while Title, Availability, Rating, and Price Category are stored as object-type variables.

The dataset dimensions and data types were inspected to ensure that the variables were represented appropriately for exploratory analysis. The Rating variable was subsequently converted into a numeric representation for correlation analysis



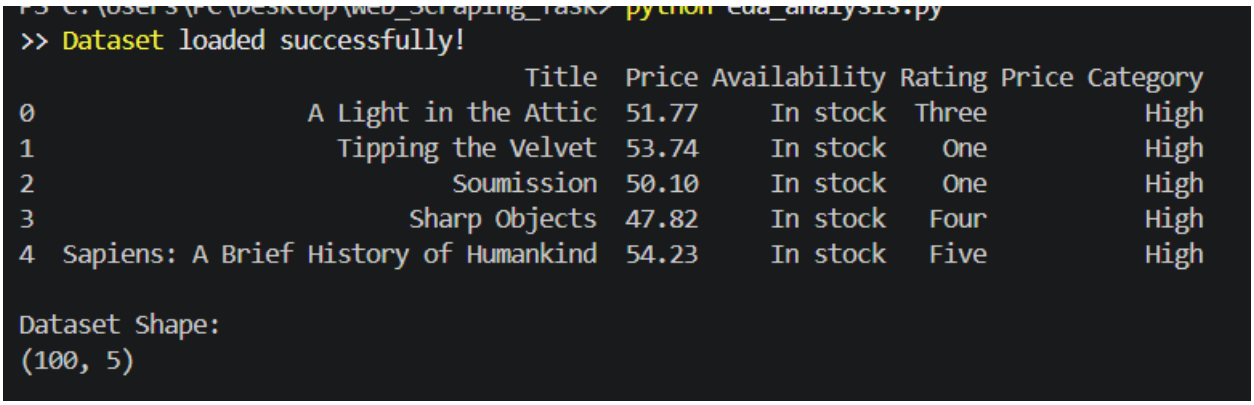


Figure 1: Dataset structure, dimensions, and data types.

6. Data Quality Assessment

Data quality was assessed by checking for missing values and duplicate records. The Pandas functions isnull().sum() and duplicated().sum() were used for this purpose.

Missing-value analysis identifies fields that do not contain recorded values, while duplicate analysis helps identify repeated rows. These checks are important because missing or duplicated data may affect the accuracy and reliability of subsequent analysis.

The results of these checks were used to evaluate the completeness and consistency of the dataset. Any detected issues should be investigated before further analysis or data-driven decision-making.

```
>> Dataset loaded successfully!
memory usage: 4.0 KB

Missing Values:
Title             0
Price             0
Availability      0
Rating            0
Price Category    0
dtype: int64
```

```
21
22    print("\nDuplicate Records:")
23    print(df.duplicated().sum())
24
```

```
Duplicate Records:
0
```

7. Statistical Analysis

Descriptive statistical analysis was performed to understand the distribution and central tendency of book prices. Pandas functions were used to calculate the mean, median, minimum, and maximum prices. The describe() function was also used to generate a statistical summary of the numerical variables.

The mean represents the average price, while the median represents the middle value when prices are arranged in ascending order. The minimum and maximum values indicate the lowest and highest observed prices, respectively.

The frequency distributions of book ratings, availability statuses, and price categories were also examined to identify the most common categories and understand the composition of the dataset.

```
Statistical Summary:
            Price
count  100.000000
mean    34.560700
std     14.638531
min     10.160000
25%     19.897500
50%     34.775000
75%     47.967500
max     58.110000
```

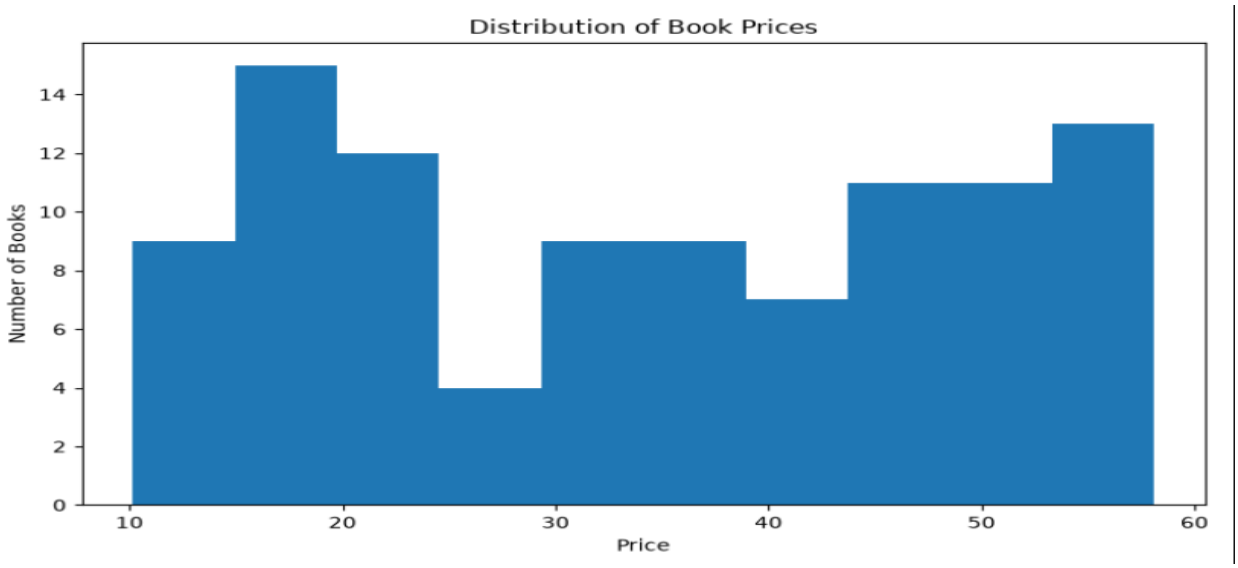
8. Trend and Pattern Analysis

The dataset was explored to identify patterns in book prices, ratings, availability, and price categories.

The rating frequency distribution was examined to identify the most common book rating. Price Category counts were used to compare the number of books classified as Low, Medium, and High priced. Availability counts were also examined to understand the recorded stock status of the collected books.

The price distribution histogram was used to investigate how book prices were spread across different price ranges. These analyses provide a descriptive overview of the collected records. However, the findings represent only the 100 books collected from the first five catalogue pages and should not automatically be generalized to all books or websites.



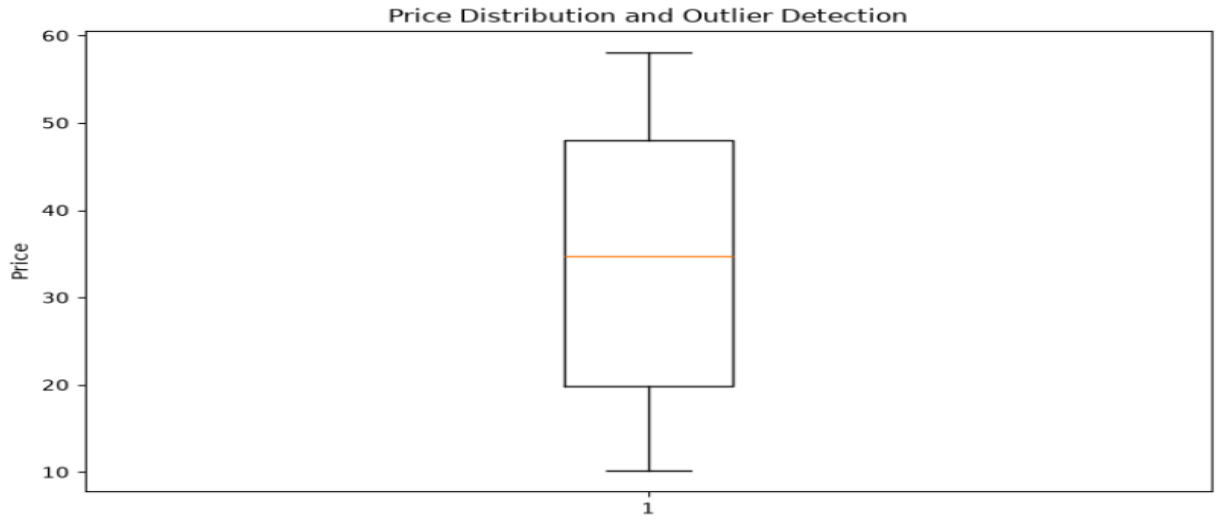


9. Outlier Detection

Potential outliers in the Price variable were investigated using a boxplot. A boxplot displays the median, interquartile range, and potential extreme observations.

The boxplot provides a visual method for identifying prices that may be unusually high or low compared with the rest of the observations. Such values should be investigated to determine whether they represent valid book prices or possible data quality problems.

A potential outlier is not necessarily an error. Since the dataset contains book prices collected from a website, an unusually expensive book may still be a valid observation. Therefore, extreme values should not be removed without an appropriate reason.



10. Hypothesis Testing and Correlation Analysis

A hypothesis was formulated to investigate whether a linear relationship exists between book prices and ratings.

Null Hypothesis (H_0): There is no linear correlation between book prices and ratings in the analyzed dataset.

Alternative Hypothesis (H_1): There is a linear correlation between book prices and ratings in the analyzed dataset.

The categorical Rating variable was mapped to a numeric scale ranging from 1 to 5. The Pearson correlation coefficient was then calculated between Price and the numeric rating.

The correlation coefficient ranges from -1 to +1. A positive value indicates a positive linear association, while a negative value indicates a negative linear association. A value near zero indicates little or no linear association.

The correlation coefficient was interpreted alongside a scatter plot. Correlation alone does not establish causation. In addition, a formal statistical significance test is required to determine whether the observed correlation provides sufficient evidence to reject the null hypothesis.



11. Data Visualization

Matplotlib was used to create visual representations of the dataset. The visualizations included a histogram of book prices, a boxplot for potential outlier detection, a bar chart showing rating frequencies, and a scatter plot examining the relationship between prices and ratings.

The histogram helps reveal the distribution of prices, while the boxplot highlights potential extreme observations. The rating bar chart compares the frequency of different rating categories, and the scatter plot helps assess whether a visible relationship exists between price and rating.

These visualizations complement the numerical analysis by making the dataset's characteristics easier to interpret.

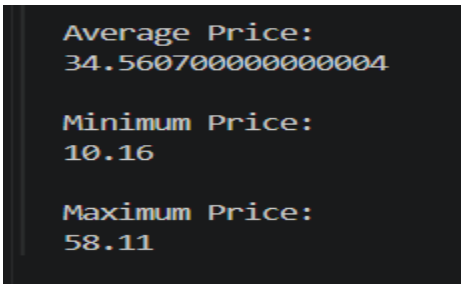
12. Findings and Results

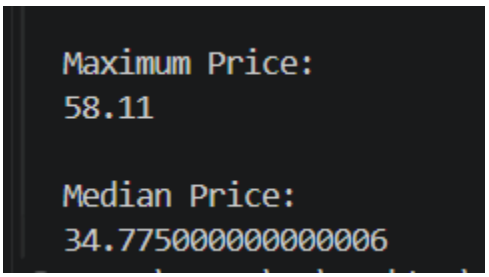
The exploratory analysis provided an overview of the structure, quality, and statistical characteristics of the collected dataset.

The dataset contains 100 book records and five original variables. Descriptive statistics were calculated to summarize book prices, while frequency counts were used to examine ratings, availability, and price categories.

Missing-value and duplicate checks were performed to assess data quality. A boxplot was used to investigate potential price outliers, and a histogram was used to examine the distribution of prices. Correlation analysis and a scatter plot were used to explore the association between book prices and ratings.

The final findings should be interpreted using the actual numerical outputs and visualizations generated during the analysis. The dataset represents a limited sample of books from five catalogue pages, so broader conclusions would require a larger and more representative dataset.





```
Rating Distribution:
Rating
Three    22
One      22
Five     19
Two      19
Four     18
Name: count, dtype: int64
```

```
Price Category Distribution:
Price Category
High      40
Medium    34
Low       26
Name: count, dtype: int64
```

```
Availability Distribution:
Availability
In stock    100
Name: count, dtype: int64
```

```
Price-Rating Correlation:
-0.12174127007413635
```

13. Conclusion

This task demonstrated the practical application of Exploratory Data Analysis using Python. The dataset was examined to understand its structure, variable types, descriptive statistics, category distributions, and potential data quality issues.

Pandas was used to inspect the dataset, calculate statistical summaries, and identify missing values and duplicate records. Matplotlib was used to visualize price distributions, rating frequencies, potential outliers, and the relationship between book prices and ratings.

The analysis established a foundation for understanding the collected data before conducting further investigation. It also highlighted the importance of validating assumptions, interpreting statistical results carefully, and investigating potential anomalies rather than automatically treating them as errors.

Overall, EDA helped organize the dataset into a form that is easier to understand and provided insights that can guide future analysis.